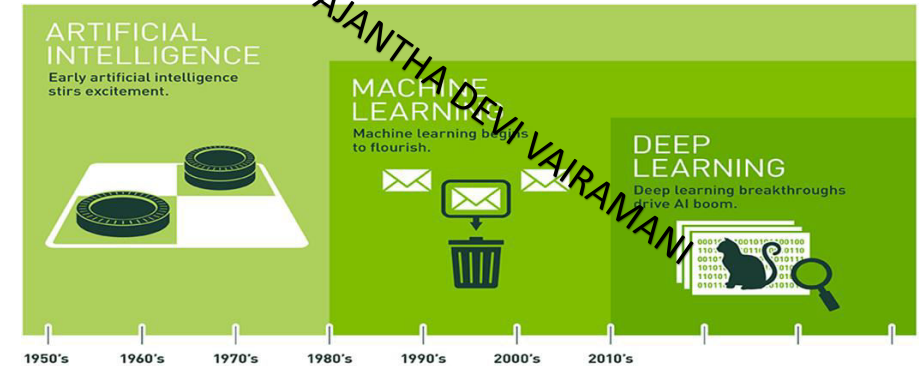


The Foundations of Machine Learning: From Data Collection to Model Evaluation

DR.V.AJANTHA DEVI
RESEARCH HEAD,

AI → Machine Learning



DR.V.AJANTHA DEVI, RESEARCH HEAD

2

How do humans make decisions?

Human decision making



DR.V.AJANTHA DEVI, RESEARCH HEAD

3

How do humans make decisions?

Human decision making



DR.V.AJANTHA DEVI, RESEARCH HEAD

4

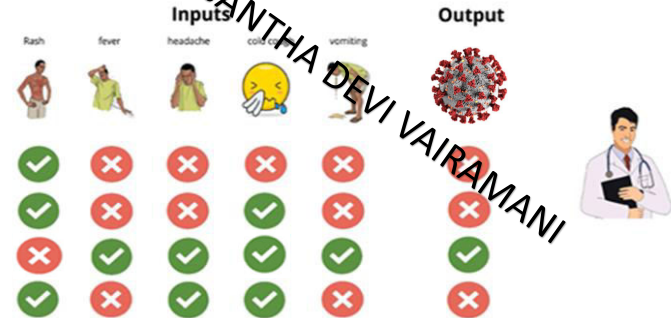
How do humans make decisions from past experiences?

Human decision making



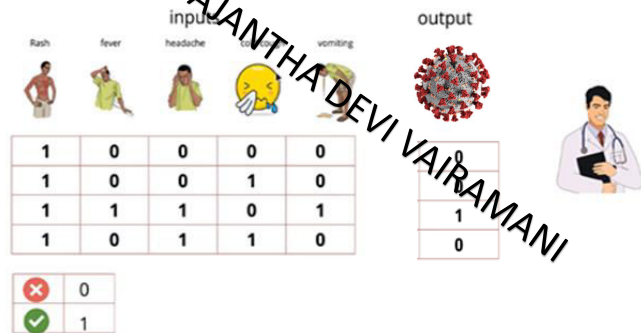
How do humans make decisions from past experiences?

Human decision making



How do humans make decisions from past experiences?

Human decision making



Is this applicable in multiple domains?

Human decision making



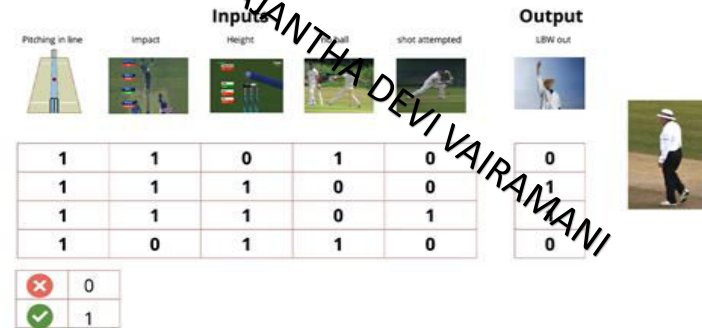
Is this applicable in multiple domains?

Human decision making



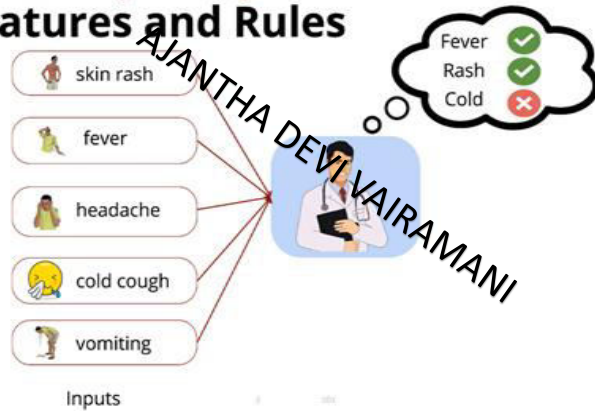
Is this applicable in multiple domains?

Human decision making



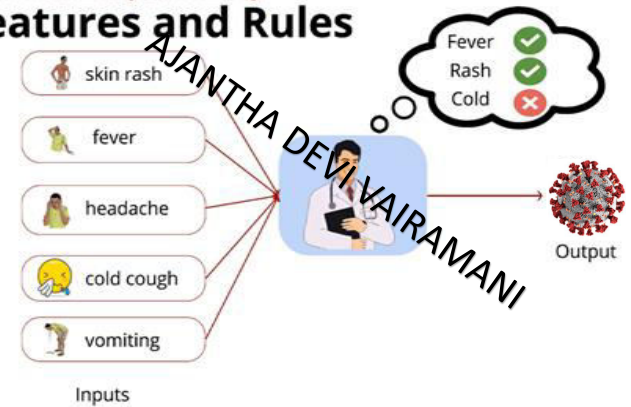
What is the semantics of decision making?

Features and Rules



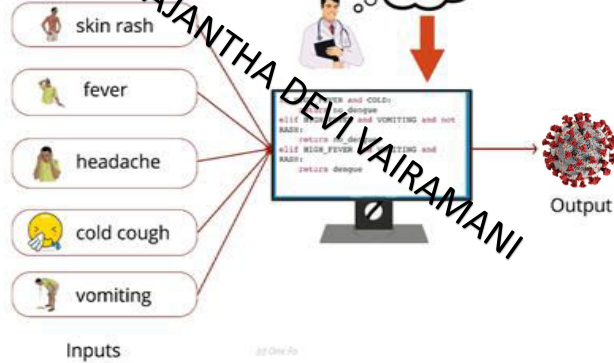
What is the semantics of decision making?

Features and Rules



How do we outsource this to a machine?

Expert Systems



Do we need to look beyond expert systems?

Limitations

Task of hiring someone for a job

- 10th marks
- 12th marks
- graduate
- CGPA
- projects

Inputs

Do we need to look beyond expert systems?

Limitations

Task of hiring someone for a job



Do we need to look beyond expert systems?

Limitations

Task of hiring someone for a job
- involving many factors

- 10th marks
- 12th marks
- graduate
- CGPA
- projects
- publications
- ...
- awards

Output

Do we need to look beyond expert systems?

Limitations

Lots of data to make sense from

10th marks	12th marks	graduate	CGPA	projects	awards
92	82	yes	9.2	3	2
83	75.2	yes	7.2	1	0
75	70	no	6.4	0	1
96	95	yes	9.5	5	4
90	89	yes	8.8	2	1
78	82	yes	7.6	0	0
86	88	yes	8.4	1	1



Do we need to look beyond expert systems?

Limitations

The rules can be very complex

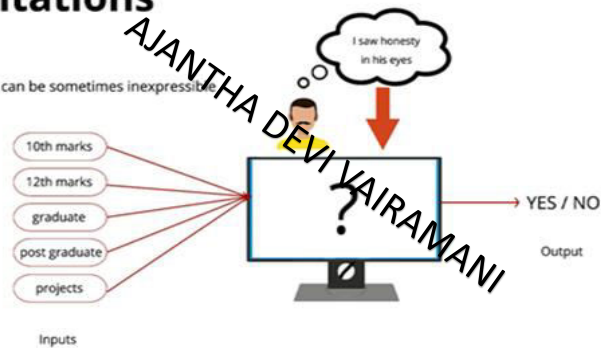
10th marks	12th marks	graduate	CGPA	projects	awards
92	82	yes	9.2	3	2
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78	82	yes	7.6	0	0
86	88	yes	8.4	1	1



Do we need to look beyond expert systems?

Limitations

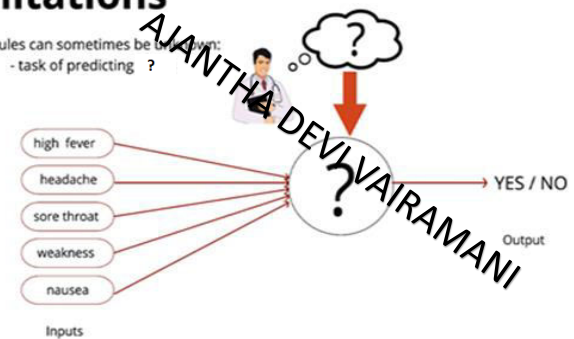
The rules can be sometimes inexpressible



Do we need to look beyond expert systems?

Limitations

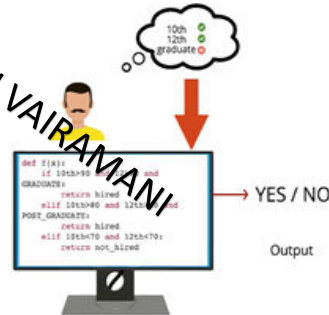
The rules can sometimes be too complex:
 - task of predicting ?



How to move from writing rules to learning rules?

Say Hi to Machine Learning

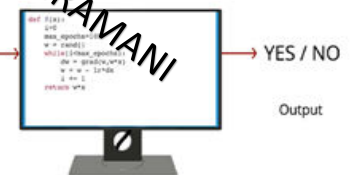
10th marks	12th marks	graduate	CGPA	projects	awards
92	82	yes	9.2	3	2
83	75.2	yes	7.2	1	0
75	70	no	6.4	0	1
96	95	yes	9.5	5	4
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How to move from writing rules to learning rules?

Say Hi to Machine Learning

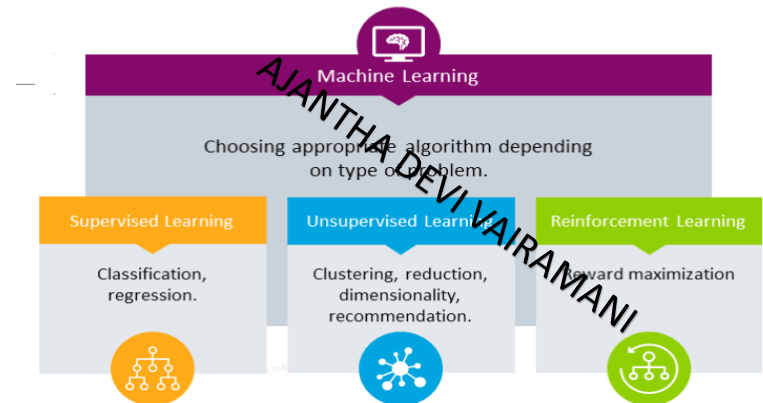
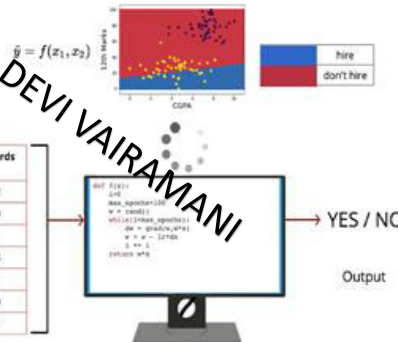
10th marks	12th marks	graduate	CGPA	projects	awards
92	82	yes	9.2	3	2
83	75.2	yes	7.2	1	0
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96	95	yes	9.5	5	4
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How to move from writing rules to learning rules?

Say Hi to Machine Learning

10th marks	12th marks	graduate	CGPA	projects	awards
92	82	yes	9.2	3	2
83	75.2	yes	7.2	1	0
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96	95	yes	9.5	5	4
90	89	yes	8.8	2	1
78	82	yes	7.6	0	0
86	88	yes	8.4	1	1



COMPONENTS OF MACHINE LEARNING



What is the fuel of Machine Learning?

Data data everywhere



What is the fuel of Machine Learning?

Data data everywhere



What is the fuel of Machine Learning?

Data data everywhere



What is the fuel of Machine Learning?

Data data everywhere



What is the fuel of Machine Learning?

Data data everywhere



What is the fuel of Machine Learning?

Data data everywhere



How do you feed data to machines ?

Data data everywhere

scans							y
	2.3	5.9	...	11.0	0.3	8.9	0
	-8.5	-1.7	...	-1.3	9.0	2.2	1
	-0.4	6.7	...	-2.4	4.7	-7.5	0
	1.6	-0.4	...	-4.6	6.4	1.9	0
	3.9	-4.1	...	6.7	-3.1	2.1	1
	5.1	3.7	...	1.8	-4.2	9.3	1

How do you feed data to machines ?

Data data everywhere

scans	x						y
	2.3	5.9	...	11.0	-0.3	0.5	0
	-8.5	-1.7	...	-1.3	9.0	7.2	1
	-0.4	6.7	...	-2.4	4.7	-7.3	0
	1.6	-0.4	...	-4.6	6.4	1.9	1
	3.9	-4.1	...	6.7	-3.1	2.1	1
	5.1	3.7	...	1.8	-4.2	9.3	1



- ✓ All data encoded as numbers
- ✓ Typically high dimensional

How do you feed data to machines ?

Data data everywhere

Document	x						y
Don't buy this Mi 6 Pro. (Speaker volume is very bad)	1.9	3.2	...	-9.8	-6.7	-2.2	negative
Delivered as shown. Good price and fits perfect	1.3	3.6	...	-5.4	9.1	2.3	positive
What a phone. A handy app phone. Mi at its best	0.4	7.6	...	-0.1	-1.4	8.7	positive
Its look stunning in pictures but not in real	1.5	-0.8	...	7.8	8.4	0.3	negative
Amazing camera and battery. Good deal!	2.5	-5.7	...	0.9	5.3	-8.1	positive

- ✓ All data encoded as numbers
- ✓ Typically high dimensional

How do you feed data to machines ?

Data data everywhere



1.3 -4.3 2.1 -6.7 ... 1.5 8.9 10.1 -4.3



2.6 7.9 -0.3 8.1 ... -4.2 0.3 1.2 9.4



-5.2 -3.2 4.2 0.3 ... 3.5 8.3 -1.4 -8.7



2.3 -5.6 -1.2 7.8 ... 9.9 10.1 -1.1 3.5



8.5 2.1 -6.3 5.3 ... 7.2 -1.3 -4.5 11.8

- ✓ All data encoded as numbers
- ✓ Typically high dimensional $\mathbb{R}^n$

What is the fuel of Machine Learning?

Data data everywhere



What do you do with this data?

Tasks



Input

Output

What do you do with this data?

Tasks

From product description to structured specifications



Input

Output

What do you do with this data?

Tasks

From specifications + reviews to writing FAQs



Input

Output

What do you do with this data?

Tasks

From specifications + reviews + FAQs to Question Answering



Input

Output

What do you do with this data?

Tasks

From specifications + reviews + personal data to recommendations



What do you do with this data?

Tasks



What do you do with this data?

Tasks

From images identify people



What do you do with this data?

Tasks

From images identify activities



What do you do with this data?

Tasks

From images identify places



Input



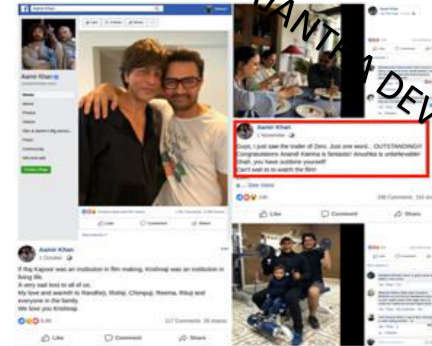
Output

Gym

What do you do with this data?

Tasks

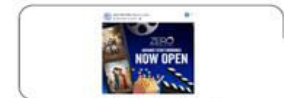
From posts recommend posts



Input



Output



What do you do with this data?

Tasks

Supervised

Classification

	x					y
	-8.5	-1.7	...	9.0	7.2	1
	-0.4	6.7	...	4.7	-7.2	0
	3.2	5.9	...	11.0	8.9	1
	2.7	3.1	...	-2.1	9.7	0
	3.9	7.8	...	-5.1	3.7	0
	7.1	0.9	...	1.5	-4.2	1

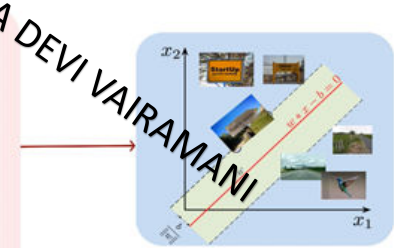
What do you do with this data?

Tasks

Supervised

Classification

	x					y
	-8.5	-1.7	...	9.0	7.2	1
	-0.4	6.7	...	4.7	-7.2	0
	3.2	5.9	...	11.0	8.9	1
	2.7	3.1	...	-2.1	9.7	0
	3.9	7.8	...	-5.1	3.7	0
	7.1	0.9	...	1.5	-4.2	1



What do you do with this data?

Tasks

Supervised

	$\mathcal{X}$					left_x	left_y	width	height
	-8.5	-1.7	...	9.0	7.2	2.3	1.2	9.2	10.1
	0.9	-2.1	...	-8.1	1.9	4.3	4.2	7.1	5.1
	2.9	-4.5	...	-3.7	8.9	2.3	7.2	6.9	7.3

Regression



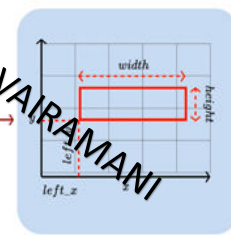
What do you do with this data?

Tasks

Supervised

	$\mathcal{X}$					left_x	left_y	width	height
	-8.5	-1.7	...	9.0	7.2	2.3	1.2	9.2	10.1
	0.9	-2.1	...	-8.1	1.9	4.3	4.2	7.1	5.1
	2.9	-4.5	...	-3.7	8.9	2.3	7.2	6.9	7.3

Regression



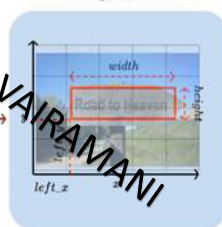
What do you do with this data?

Tasks

Supervised

	$\mathcal{X}$					left_x	left_y	width	height
	-8.5	-1.7	...	9.0	7.2	2.3	1.2	9.2	10.1
	0.9	-2.1	...	-8.1	1.9	4.3	4.2	7.1	5.1
	2.9	-4.5	...	-3.7	8.9	2.3	7.2	6.9	7.3

Regression



What do you do with this data?

Tasks

Un Supervised

	$\mathcal{X}$				
	-8.5	-1.7	...	9.0	7.2
	-0.4	6.7	...	4.7	-4.1
	3.2	5.9	...	11.0	8.9
	2.7	3.1	...	-2.1	9.7
	3.9	7.8	...	-5.1	3.7

Clustering



What do you do with this data?

Tasks

UnSupervised

	$\mathcal{X}$				
	-8.5	-1.7	...	9.0	7.2
	-0.4	6.7	...	4.7	-4.1
	3.2	5.9	...	11.0	8.9
	2.7	3.1	...	-2.1	9.7
	3.9	7.8	...	-5.1	3.7

Clustering



What do you do with this data?

Tasks

UnSupervised

	$\mathcal{X}$				
	-8.5	-1.7	...	9.0	7.2
	-0.4	6.7	...	4.7	-4.1
	3.2	5.9	...	11.0	8.9
	2.7	3.1	...	-2.1	9.7
	3.9	7.8	...	-5.1	3.7

Clustering



What do you do with this data?

Tasks

UnSupervised

	$\mathcal{X}$				
	-8.5	-1.7	...	9.0	7.2
	-0.4	6.7	...	4.7	-4.1
	3.2	5.9	...	11.0	8.9
	2.7	3.1	...	-2.1	9.7
	3.9	7.8	...	-5.1	3.7

Generation



What do you do with this data?

Tasks

UnSupervised

	$\mathcal{X}$				
	-8.5	-1.7	...	9.0	7.2
	-0.4	6.7	...	4.7	-4.1
	3.2	5.9	...	11.0	8.9
	2.7	3.1	...	-2.1	9.7
	3.9	7.8	...	-5.1	3.7

Generation



What do you do with this data?

Tasks

UnSupervised

Generation

Tweets	x					
Wow, but not surprisingly, ObamaCare was just ruled UNCONSTITUTIONAL, by a highly respected judge in Texas. Great news for America!	2.3	5.9	...	11.0	-0.3	...
For the record, there were MANY people who wanted to be the White House Chief of Staff. Much to well do a GREAT job!	-8.5	-1.7	...	-1.3	9.0	7.2
FAKE NEWS - THE ENEMY OF THE PEOPLE!	-0.4	6.7	...	-2.4	4.7	-6.2
The Democrats and President Obama gave me 150 Billion Dollars and got nothing, but they can't give 5 Billion Dollars for National Security and a war?	1.6	-0.4	...	-4.6	6.4	1.9

What do you do with this data?

Tasks

UnSupervised

Generation

Tweets	x					
Wow, but not surprisingly, ObamaCare was just ruled UNCONSTITUTIONAL, by a highly respected judge in Texas. Great news for America!	2.3	5.9	...	11.0	-0.3	...
For the record, there were MANY people who wanted to be the White House Chief of Staff. Much to well do a GREAT job!	-8.5	-1.7	...	-1.3	9.0	7.2
FAKE NEWS - THE ENEMY OF THE PEOPLE!	-0.4	6.7	...	-2.4	4.7	-6.2
The Democrats and President Obama gave me 150 Billion Dollars and got nothing, but they can't give 5 Billion Dollars for National Security and a war?	1.6	-0.4	...	-4.6	6.4	1.9



What do you do with this data?

Tasks



What is the mathematical formulation of a task?

Models

x

- ☐ ba
- ☐ car
- ☐ dog
- ☐ cat
- ☐ ship

What is the mathematical formulation of a task?

Models

x



$\begin{bmatrix} 2.1, 1.2, \dots, 5.6, 7.2 \\ 0.1, 3.1, \dots, 1.7, 3.4 \\ 0.5, 9.1, \dots, 5.1, 0.8 \\ 1.2, 4.1, \dots, 6.3, 7.4 \end{bmatrix}$

- ☒ bat
- ☐ car
- ☐ dog
- ☐ cat
- ☐ ship

$\begin{bmatrix} 0, 0, 1, 0, 0 \\ 0, 1, 0, 0, 0 \\ 0, 0, 0, 0, 1 \\ 1, 0, 0, 0, 0 \end{bmatrix}$

$$y = f(x) \text{ [true relation, unknown]}$$

What is the mathematical formulation of a task?

Models

x



$\begin{bmatrix} 2.1, 1.2, \dots, 5.6, 7.2 \\ 0.1, 3.1, \dots, 1.7, 3.4 \end{bmatrix}$

- ☐ bat
- ☒ car
- ☐ dog
- ☐ cat
- ☐ ship

$\begin{bmatrix} 0, 0, 1, 0, 0 \\ 0, 1, 0, 0, 0 \end{bmatrix}$

$$y = f(x) \text{ [true relation, unknown]}$$

What is the mathematical formulation of a task?

Models

x



$\begin{bmatrix} 2.1, 1.2, \dots, 5.6, 7.2 \end{bmatrix}$

- ☐ bat
- ☐ car
- ☒ dog
- ☐ cat
- ☐ ship

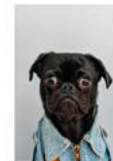
$\begin{bmatrix} 0, 0, 1, 0, 0 \end{bmatrix}$

$$y = f(x) \text{ [true relation, unknown]}$$

What is the mathematical formulation of a task?

Models

x



$\begin{bmatrix} 2.1, 1.2, \dots, 5.6, 7.2 \\ 0.1, 3.1, \dots, 1.7, 3.4 \\ 0.5, 9.1, \dots, 5.1, 0.8 \\ 1.2, 4.1, \dots, 6.3, 7.4 \\ 3.2, 2.1, \dots, 3.1, 0.9 \end{bmatrix}$

- ☐ bat
- ☐ car
- ☒ dog
- ☐ cat
- ☐ ship

$\begin{bmatrix} 0, 0, 1, 0, 0 \\ 0, 1, 0, 0, 0 \\ 0, 0, 0, 0, 1 \\ 1, 0, 0, 0, 0 \\ 0, 0, 1, 0, 0 \end{bmatrix}$

$$y = f(x) \text{ [true relation, unknown]}$$

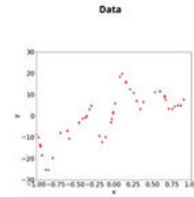
$$\hat{y} = \hat{f}(x) \text{ [our approximation]}$$

What are the choices for $\hat{f}$?

Models

$$\hat{y} = \hat{f}(x) \text{ [our approximation]}$$

x	y
0.5	14.8
0.2	13.3
0.6	11.6
...	...
0.3	6.16

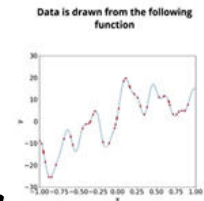


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x	y
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0.6	11.6
...	...
0.3	6.16



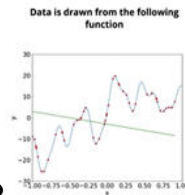
What are the choices for $\hat{f}$?

Models

$$\hat{y} = \hat{f}(x) \text{ [our approximation]}$$

$$\hat{y} = mx + c$$

x	y
0.5	14.8
0.2	13.3
0.6	11.6
...	...
0.3	6.16



What are the choices for $\hat{f}$?

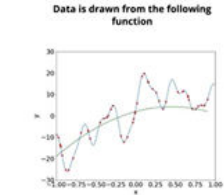
Models

$$\hat{y} = \hat{f}(x) \text{ [our approximation]}$$

$$\hat{y} = mx + c$$

$$\hat{y} = ax^2 + bx + c$$

x	y
0.5	14.8
0.2	13.3
0.6	11.6
...	...
0.3	6.16



What are the choices for f ?

Models

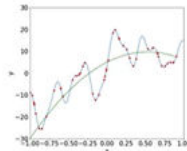
$$\hat{y} = \hat{f}(x) \text{ [our approximation]}$$

$$\hat{y} = mx + c$$

$$\hat{y} = ax^2 + bx + c$$

x	y
0.5	14.8
0.2	13.3
0.1	11.6
...	...
0.3	6.16

Data is drawn from the following function



What are the choices for f ?

Models

$$\hat{y} = \hat{f}(x) \text{ [our approximation]}$$

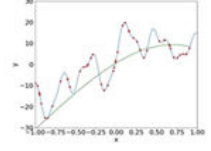
$$\hat{y} = mx + c$$

$$\hat{y} = ax^2 + bx + c$$

$$\hat{y} = ax^3 + bx^2 + cx + d$$

x	y
0.5	14.8
0.2	13.3
0.1	11.6
...	...
0.3	6.16

Data is drawn from the following function



What are the choices for f ?

Models

$$\hat{y} = \hat{f}(x) \text{ [our approximation]}$$

$$\hat{y} = mx + c$$

$$\hat{y} = ax^2 + bx + c$$

$$\hat{y} = ax^3 + bx^2 + cx + d$$

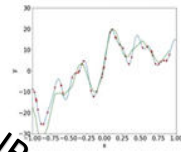
$$\hat{y} = ax^4 + bx^3 + cx + d$$

...

$$\hat{y} = ax^{25} + bx^{24} + \dots + cx + d$$

x	y
0.5	14.8
0.2	13.3
0.1	11.6
...	...
0.3	6.16

Data is drawn from the following function



What are the choices for f ?

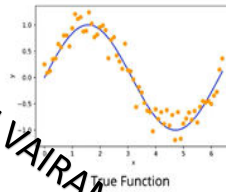
Models



How do we know which model is better ?

Loss Function

	x	y
1	0.00	0.24
2	0.10	0.08
3	0.20	0.12
....		
n	6.40	0.36

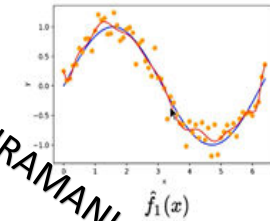


True Function

How do we know which model is better ?

Loss Function

	x	y	$\hat{f}_1(x)$
1	0.00	0.24	0.25
2	0.10	0.08	0.09
3	0.20	0.12	0.11
....			
n	6.40	0.36	0.36



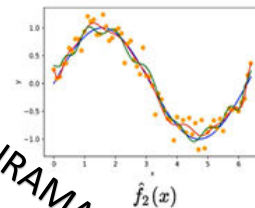
$\hat{f}_1(x)$

 $\hat{f}_1(x) = 1.79x^{25} - 4.54x^{24} + \dots - 1.48x + 2.48$

How do we know which model is better ?

Loss Function

	x	y	$\hat{f}_1(x)$	$\hat{f}_2(x)$
1	0.00	0.24	0.25	0.32
2	0.10	0.08	0.09	0.30
3	0.20	0.12	0.11	0.31
....				
n	6.40	0.36	0.36	0.22



$\hat{f}_2(x)$

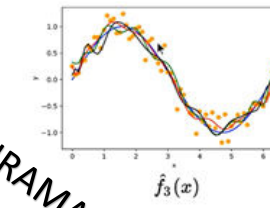
 $\hat{f}_1(x) = 1.79x^{25} - 4.54x^{24} + \dots - 1.48x + 2.48$

 $\hat{f}_2(x) = 2.27x^{25} + 9.89x^{24} + \dots + 2.79x + 3.22$

How do we know which model is better ?

Loss Function

	x	y	$\hat{f}_1(x)$	$\hat{f}_2(x)$	$\hat{f}_3(x)$
1	0.00	0.24	0.25	0.32	0.08
2	0.10	0.08	0.09	0.30	0.20
3	0.20	0.12	0.11	0.31	0.31
....					
n	6.40	0.36	0.36	0.22	0.15



$\hat{f}_3(x)$

 $\hat{f}_1(x) = 1.79x^{25} - 4.54x^{24} + \dots - 1.48x + 2.48$

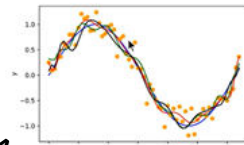
 $\hat{f}_2(x) = 2.27x^{25} + 9.89x^{24} + \dots + 2.79x + 3.22$

 $\hat{f}_3(x) = 3.78x^{25} + 1.57x^{24} + \dots + 1.01x + 8.68$

How do we know which model is better ?

Loss Function

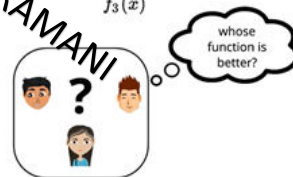
	x	y	$\hat{f}_1(x)$	$\hat{f}_2(x)$	$\hat{f}_3(x)$
1	0.00	0.24	0.25	0.32	0.08
2	0.10	0.08	0.09	0.30	0.20
3	0.20	0.12	0.11	0.31	0.15
...	...	...	...	...	...
n	6.40	0.36	0.36	0.22	0.15

 $\hat{f}_3(x)$

$$\hat{f}_1(x) = 1.79x^{25} - 4.54x^{24} + \dots - 1.48x + 2.48$$

$$\hat{f}_2(x) = 2.27x^{25} + 9.89x^{24} + \dots + 2.79x + 3.22$$

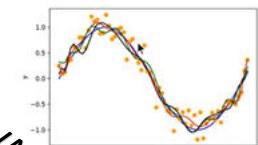
$$\hat{f}_3(x) = 3.78x^{25} + 1.57x^{24} + \dots + 1.01x + 8.68$$



How do we know which model is better ?

Loss Function

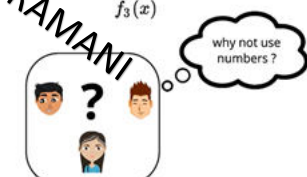
	x	y	$\hat{f}_1(x)$	$\hat{f}_2(x)$	$\hat{f}_3(x)$
1	0.00	0.24	0.25	0.32	0.08
2	0.10	0.08	0.09	0.30	0.20
3	0.20	0.12	0.11	0.31	0.15
...	...	...	...	...	...
n	6.40	0.36	0.36	0.22	0.15

 $\hat{f}_3(x)$

$$\hat{f}_1(x) = 1.79x^{25} - 4.54x^{24} + \dots - 1.48x + 2.48$$

$$\hat{f}_2(x) = 2.27x^{25} + 9.89x^{24} + \dots + 2.79x + 3.22$$

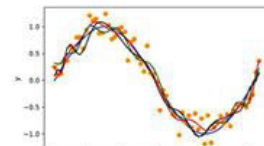
$$\hat{f}_3(x) = 3.78x^{25} + 1.57x^{24} + \dots + 1.01x + 8.68$$



How do we know which model is better ?

Loss Function

	x	y	$\hat{f}_1(x)$	$\hat{f}_2(x)$	$\hat{f}_3(x)$
1	0.00	0.24	0.25	0.32	0.08
2	0.10	0.08	0.09	0.30	0.20
3	0.20	0.12	0.11	0.31	0.15
...	...	...	...	...	...
n	6.40	0.36	0.36	0.22	0.15

 $\hat{f}_3(x)$

$$\hat{f}_1(x) = 1.79x^{25} - 4.54x^{24} + \dots - 1.48x + 2.48$$

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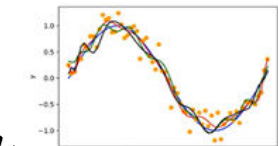
$$\hat{f}_3(x) = 3.78x^{25} + 1.57x^{24} + \dots + 1.01x + 8.68$$

$$\mathcal{L}_1 = \sum_{i=1}^n (y_i - \hat{f}_1(x_i))^2$$

How do we know which model is better ?

Loss Function

	x	y	$\hat{f}_1(x)$	$\hat{f}_2(x)$	$\hat{f}_3(x)$
1	0.00	0.24	0.25	0.32	0.08
2	0.10	0.08	0.09	0.30	0.20
3	0.20	0.12	0.11	0.31	0.15
...	...	...	...	...	...
n	6.40	0.36	0.36	0.22	0.15

 $\hat{f}_3(x)$

$$\hat{f}_1(x) = 1.79x^{25} - 4.54x^{24} + \dots - 1.48x + 2.48$$

$$\hat{f}_2(x) = 2.27x^{25} + 9.89x^{24} + \dots + 2.79x + 3.22$$

$$\hat{f}_3(x) = 3.78x^{25} + 1.57x^{24} + \dots + 1.01x + 8.68$$

$$\mathcal{L}_1 = \sum_{i=1}^n (y_i - \hat{f}_1(x_i))^2$$

$$\mathcal{L}_2 = \sum_{i=1}^n (y_i - \hat{f}_2(x_i))^2$$

$$\mathcal{L}_3 = \sum_{i=1}^n (y_i - \hat{f}_3(x_i))^2$$

How do we know which model is better ?

Loss Function

	x	y	$\hat{f}_1(x)$	$\hat{f}_2(x)$	$\hat{f}_3(x)$	
1	0.00	0.24	0.25	0.32	0.08	$\mathcal{L}_1 = \sum_{i=1}^n (y_i - \hat{f}_1(x_i))^2$ $= (0.24 - 0.25)^2 + (0.08 - 0.09)^2 + (0.12 - 0.11)^2 + \dots + (0.36 - 0.36)^2$ $= 1.38$
2	0.10	0.08	0.09	0.30	0.12	
3	0.20	0.12	0.11	0.31	0.15	
...	...	...	...	...	...	
n	6.40	0.36	0.36	0.22	0.15	

 $\mathcal{L}_1 = \sum_{i=1}^n (y_i - \hat{f}_1(x_i))^2 = 1.38$

 $\mathcal{L}_2 = \sum_{i=1}^n (y_i - \hat{f}_2(x_i))^2 = 2.02$

 $\mathcal{L}_3 = \sum_{i=1}^n (y_i - \hat{f}_3(x_i))^2 = 2.34$

How do we know which model is better ?

Loss Function

	x	y	$\hat{f}_1(x)$	$\hat{f}_2(x)$	$\hat{f}_3(x)$	
1	0.00	0.24	0.25	0.32	0.08	$\mathcal{L}_1 = \sum_{i=1}^n (y_i - \hat{f}_1(x_i))^2$ $= (0.24 - 0.25)^2 + (0.08 - 0.09)^2 + (0.12 - 0.11)^2 + \dots + (0.36 - 0.36)^2$ $= 1.38$
2	0.10	0.08	0.09	0.30	0.12	
3	0.20	0.12	0.11	0.31	0.15	
...	...	...	...	...	...	
n	6.40	0.36	0.36	0.22	0.15	

 $\mathcal{L}_1 = \sum_{i=1}^n (y_i - \hat{f}_1(x_i))^2 = 1.38$

 $\mathcal{L}_2 = \sum_{i=1}^n (y_i - \hat{f}_2(x_i))^2 = 2.02$

 $\mathcal{L}_3 = \sum_{i=1}^n (y_i - \hat{f}_3(x_i))^2 = 2.34$

- ✓ Squared Error Loss
- ✓ Cross Entropy Loss
- ✓ KL divergence

What does a loss function look like ?

Loss Function



How do we identify parameters of the model?

Learning Algorithm

$$f(x) = ax_1^2 + bx_2^3 + cx_3^2$$

Budget in (100crs)	Box Office Collection in (100 crs)	Action Scene in times (100 mins)	IMDB Rating
0.55	0.66	0.22	4.8
0.68	0.91	0.77	7.2
0.66	0.88	0.67	6.7
...	...	...	...
0.72	0.94	0.97	8.1
0.58	0.74	0.35	5.3

How do we identify parameters of the model?

Learning Algorithm

Budget in (100crs)	Box Office Collection in (100 crs)	Action Scene in times (100 mins)	IMDB Rating
0.55	0.66	0.22	4.8
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0.66	0.88	0.67	6.7
0.72	0.94	0.97	8.1
0.58	0.74	0.35	5.3

$$f(x) = ax_1^2 + bx_2^3 + cx_3^2$$

$$\hat{f}(x) = 3.5x_1^2 + 2.5x_2^3 + 1.2x_3^2$$

How do we identify parameters of the model?

Learning Algorithm

Budget in (100crs)	Box Office Collection in (100 crs)	Action Scene in times (100 mins)	IMDB Rating
0.55	0.66	0.22	4.8
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0.66	0.88	0.67	6.7
0.72	0.94	0.97	8.1
0.58	0.74	0.35	5.3

$$f(x) = ax_1^2 + bx_2^3 + cx_3^2$$

$$\mathcal{L}_1 = \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$

$$\hat{f}(x) = 3.5x_1^2 + 2.5x_2^3 + 1.2x_3^2$$

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$$\mathcal{L}_1 = \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$

$$\hat{f}(x) = 3.5x_1^2 + 2.5x_2^3 + 1.2x_3^2$$

How do we identify parameters of the model?

Learning Algorithm

Budget in (100crs)	Box Office Collection in (100 crs)	Action Scene times (100 mins)	IMDB Rating
0.55	0.66	0.22	4.8
0.68	0.91	0.77	7.2
0.66	0.88	0.67	6.7
⋮	⋮	⋮	⋮
0.72	0.94	0.97	8.1
0.58	0.74	0.35	5.3

$$\hat{f}(x) = ax_1^2 + bx_2^3 + cx_3^2$$

a	b	c	Loss	Min Loss
10.0	10.0	0.0	7.82	7.82

$$\mathcal{L}_1 = \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$

$$\hat{f}(x) = 3.5x_1^2 + 2.5x_2^3 + 1.2x_3^2$$

How do you formulate this mathematically?

Learning Algorithm

Budget (100crore)	Box Office Collection(100 crore)	Action Scene times (100 mins)	IMDB Rating
0.55	0.66	0.22	4.8
0.68	0.91	0.77	7.2
0.66	0.88	0.67	6.7
⋮	⋮	⋮	⋮
0.72	0.94	0.97	8.1
0.58	0.74	0.35	5.3

$$\hat{f}(x) = ax_1^2 + bx_2^3 + cx_3^2$$

a	b	c	Loss	Min Loss
20.0	20.0	-10.0	3.74	3.74

$$\mathcal{L}_1 = \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$

✗ In practice, brute force search is infeasible

How do you formulate this mathematically?

Learning Algorithm

Budget (100crore)	Box Office Collection(100 crore)	Action Scene times (100 mins)	IMDB Rating
0.55	0.66	0.22	4.8
0.68	0.91	0.77	7.2
0.66	0.88	0.67	6.7
⋮	⋮	⋮	⋮
0.72	0.94	0.97	8.1
0.58	0.74	0.35	5.3

$$\hat{f}(x) = ax_1^2 + bx_2^3 + cx_3^2$$

a	b	c	Loss	Min Loss
20.0	20.0	-10.0	3.74	3.74

Find a, b, c such that $\mathcal{L}_1 = \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$ is minimized

✗ In practice, brute force search is infeasible

How do you formulate this mathematically?

Learning Algorithm

Budget (100crore)	Box Office Collection(100 crore)	Action Scene times (100 mins)	IMDB Rating
0.55	0.66	0.22	4.8
0.68	0.91	0.77	7.2
0.66	0.88	0.67	6.7
⋮	⋮	⋮	⋮
0.72	0.94	0.97	8.1
0.58	0.74	0.35	5.3

$$\hat{f}(x) = ax_1^2 + bx_2^3 + cx_3^2$$

a	b	c	Loss	Min Loss
20.0	20.0	-10.0	3.74	3.74

$$\min_{a,b,c} \mathcal{L}_1 = \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$

✓ Many optimization solvers are available

How do you formulate this mathematically?

Learning Algorithm

Budget (100crore)	Box Office Collection(100 crore)	Action Scene times (100 mins)	IMDB Rating
0.55	0.66	0.22	4.8
0.68	0.91	0.77	7.2
0.66	0.88	0.67	6.7
⋮	⋮	⋮	⋮
0.72	0.94	0.97	8.1
0.58	0.74	0.35	5.3

$$\hat{f}(x) = ax_1^2 + bx_2^3 + cx_3^2$$

Gradient Descent ++

Adagrad

AdaProp

Adam

Many optimization solvers are available

$$\min_{a,b,c} \mathcal{L}_1 = \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$

How do you formulate this mathematically?

Learning Algorithm



How do we compute a score for our ML model?

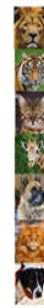
Evaluation



Class Labels	Id
Lion	1
Tiger	2
Cat	3
Giraffe	4
Dog	5

How do we compute a score for our ML model?

Evaluation



[2.1, 1.2, ..., 5.6, 7.8]
 [3.5, 6.6, ..., 2.5, 6.3]
 [6.3, 2.6, ..., 4.5, 3.8]
 [2.8, 3.6, ..., 7.5, 2.1]
 [2.2, 1.7, ..., 2.5, 1.8]
 [6.3, 2.6, ..., 4.5, 3.8]
 [1.9, 3.3, ..., 4.2, 1.1]

Class Labels	Id
Lion	1
Tiger	2
Cat	3
Giraffe	4
Dog	5

How do we compute a score for our ML model?

Evaluation

	True Labels	Predicted Labels		Class Labels	Id
	[2.1, 1.2, ..., 5.6, 7.8]	1	1	Lion	1
	[3.5, 6.6, ..., 2.5, 6.3]	2	3	Tiger	2
	[6.3, 2.6, ..., 4.5, 3.8]	3	1	Cat	3
	[2.8, 3.6, ..., 7.5, 2.1]	4	4	Giraffe	4
	[2.2, 1.7, ..., 2.5, 1.8]	5	5	Dog	5
	[6.3, 2.6, ..., 4.5, 3.8]	3	2		
	[1.9, 3.3, ..., 4.2, 1.1]	5	5		

How do we compute a score for our ML model?

Evaluation

	True Labels	Predicted Labels	Top - 1		Class Labels	Id
	[2.1, 1.2, ..., 5.6, 7.8]	1	1	✓	Lion	1
	[3.5, 6.6, ..., 2.5, 6.3]	2	3	✗	Tiger	2
	[6.3, 2.6, ..., 4.5, 3.8]	3	1	✗	Cat	3
	[2.8, 3.6, ..., 7.5, 2.1]	4	4	✓	Giraffe	4
	[2.2, 1.7, ..., 2.5, 1.8]	5	5	✓	Dog	5
	[6.3, 2.6, ..., 4.5, 3.8]	3	2	✗		
	[1.9, 3.3, ..., 4.2, 1.1]	5	5	✓		

How do we compute a score for our ML model?

Evaluation

	True Labels	Predicted Labels	Top - 1		Class Labels	Id
	[2.1, 1.2, ..., 5.6, 7.8]	1	✓		Lion	1
	[3.5, 6.6, ..., 2.5, 6.3]	2	3	✗	Tiger	2
	[6.3, 2.6, ..., 4.5, 3.8]	3	1	✗	Cat	3
	[2.8, 3.6, ..., 7.5, 2.1]	4	4	✓	Giraffe	4
	[2.2, 1.7, ..., 2.5, 1.8]	5	5	✓	Dog	5
	[6.3, 2.6, ..., 4.5, 3.8]	3	2	✗		
	[1.9, 3.3, ..., 4.2, 1.1]	5	5	✓		

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}} = \frac{4}{7} = 0.55$$

How do we compute a score for our ML model?

Evaluation

	True Labels	Predicted Labels		Class Labels	Id
	[2.1, 1.2, ..., 5.6, 7.8]	1	[1, 2, 3]	Lion	1
	[3.5, 6.6, ..., 2.5, 6.3]	2	[1, 5, 1]	Tiger	2
	[6.3, 2.6, ..., 4.5, 3.8]	3	[1, 2, 3]	Giraffe	4
	[2.8, 3.6, ..., 7.5, 2.1]	4	[4, 5, 3]	Dog	5
	[2.2, 1.7, ..., 2.5, 1.8]	5	[5, 2, 1]		
	[6.3, 2.6, ..., 4.5, 3.8]	3	[2, 1, 4]		
	[1.9, 3.3, ..., 4.2, 1.1]	5	[5, 4, 1]		

How do we compute a score for our ML model?

Evaluation

Top - 3		True Labels		Predicted Labels	
	[2.1, 1.2, ..., 5.6, 7.8]	1	[1, 4, 3]	✓	
	[3.5, 6.6, ..., 2.5, 6.3]	2	[1, 2, 3]	✓	
	[6.3, 2.6, ..., 4.5, 3.8]	3	[1, 2, 3]	✓	
	[2.8, 3.6, ..., 7.5, 2.1]	4	[4, 5, 3]	✓	
	[2.2, 1.7, ..., 2.5, 1.8]	5	[5, 2, 1]	✓	
	[6.3, 2.6, ..., 4.5, 3.8]	3	[2, 1, 4]	✗	
	[1.9, 3.3, ..., 4.2, 1.1]	5	[5, 4, 1]	✓	

$$\text{Accuracy} = \frac{\text{Number of correct predictions in top-3}}{\text{Total number of predictions}} = \frac{6}{7} = 0.86$$

Class Labels	Id
Lion	1
Tiger	2
Cat	3
Giraffe	4
Dog	5

Should we learn and test on the same data?

Evaluation

Training Data		Test Data	
x	y	x	y
	[2.1, 1.2, ..., 5.6, 7.8]	1	1
	[3.5, 6.6, ..., 2.5, 6.3]	2	2
	[6.3, 2.6, ..., 4.5, 3.8]	3	3
	[2.8, 3.6, ..., 7.5, 2.1]	4	4
	[2.2, 1.7, ..., 2.5, 1.8]	2	2

$$\hat{f}_1(x) = ax_1^2 + bx_2^3 + cx_3^2$$

$$\min_{a,b,c} \mathcal{L}_1 = \sum_{i=1}^n (y_i - \hat{f}_1(x_i))^2$$

Should we learn and test on the same data?

Evaluation

Training Data		Test Data	
x	y	x	y
	[2.1, 1.2, ..., 5.6, 7.8]	1	1
	[3.5, 6.6, ..., 2.5, 6.3]	2	2
	[6.3, 2.6, ..., 4.5, 3.8]	3	3
	[2.8, 3.6, ..., 7.5, 2.1]	4	4
	[2.2, 1.7, ..., 2.5, 1.8]	2	2

$$\hat{f}_1(x) = ax_1^2 + bx_2^3 + cx_3^2$$

$$\min_{a,b,c} \mathcal{L}_1 = \sum_{i=1}^n (y_i - \hat{f}_1(x_i))^2$$

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}$$

Putting it all together



